ELECT MAINT Department
SIMHADRI

Note No :1

Ref:SMTPP/ELECT MAINT/2023-24/EMD-CHP/374239

Date: 07/08/2023(18:23:54)

Subject:LMI revision High current bus bar system connected to generator and generator transformer (LMI/OGN/OPS/ELEC/035)

LMI of High current bus bar system connected to generator and generator transformer (LMI/OGN/OPS/ELEC/035) is revised based on latest OGN (COS-ISO-00-OGN/OPS/ELEC/035 Rev. No.: 1 Aug. 2022)

Both revised LMI and OGN are attached herewith.

Put up for approval please.

Submitted by:

मिलनकुमार माझी - वरिष्ठ प्रबंधक

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Note No:2

Put up for approval plz.

Submitted by:

मकाला नागेश्वर राव - उप महाप्रबंधक

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Note No:3

Reviewed and found in line with OGN/ELEC/35.

Please route through all sectional incharges of EMD for comprehensive Review.

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

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Note No:4

Routed through HT/Transformer, Generator and Switchyard group for review.

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Note No:5

Reviewed and found in line with OGN.

Submitted by:

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Note No:6

Reviewed and found in line with OGN.

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Note No:7

checked and found in line with OGN

Submitted by:

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Note No:8

Reviewe and found in order in line with OGN-OPS-ELEC-035_Rev1_High Current Bus Bar System Connected to generators and Generator Transformers.

May Approve.

Submitted by:

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Note No:9

Reviewed and found in order

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Note No:11

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Note No:12

May please be accorded approval.

Submitted by:

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Note No:13

Pl review the LMI w.r.to the OD and resposnibilities of Operation Dept mentioned in the LMI (if any) may be noted and forward for further approval

Submitted by:

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Note No:14

Reviewed and found no responsibility assigned to Operation Dept in OGN.

Cluase No. 7.1 & 7.2 are same and repeating in LMI.

Submitted by:

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Note No:15

Reviewed and Forwarded for further approval.

Initiator may see the comments on the pre page before forwarding further

Submitted by:

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Note No:16

Complied.

Submitted by:

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Note No:17

Reviewed.

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Note No:18

Reviewed and forwarded for futher approval

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Note No:19

Forwarded for approval please.

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Note No:20

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Note No:21

Pl examine.

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Note No:22

Examined the proposed LMI in annexure and found in with the approved OGN (COS-ISO-00-OGN/OPS/ELEC/035 Rev. No.: 1 Aug. 2022).

Submitted for further processing.

Submitted by:

आशित रॉय चौधुरी - वरिष्ठ प्रबंधक

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Note No:23

Submitted for approval.

Submitted by:

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Note No:24

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Note No:25

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

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On:15/09/2023(09:20:40) **Ref:**SMTPP/ELECT MAINT/2023-24/EMD-CHP/374239

NTPC Limited

SIMHADRI SUPER THERMAL POWER PROJECT

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/ELEC/035

Rev. No 4

Aug 2023

High current bus bar system connected to generator and generator transformer

Online Approval for Implementation by:

Sanjay Kumar Sinha, HOP (SIMHADRI)

E Office Ref No.: SMTPP/ELECT MAINT-/2023-24/EMD CHP/ 374239

Contents

1.0 INTRODUCTION

Braided flexible connections, laminated connections, and bolted joints between the terminals of generators, generator transformers and high current busbar systems can deteriorate due to several factors. These, as a general rule, are vibrations, overheating of joints and damage caused during removal and replacement of various parts during maintenance outages. Problems have also been experienced with poor current distribution at braided connections, resulting in braids overheating. The majority of the latter problems have been addressed by redesign or altering `braid formations and new installations should have improved arrangements at interfaces with generators, transformers or generator neutral end terminations.

When major modifications have been made, the temperature of joints or braid connections should be monitored using infra-red camera techniques.

Original soldered terminal braids were not always manufactured or installed to the required high standards, often having poor soldering, distorted contact surfaces, cramped or tight braiding and braid distortion due to misalignment.

Contact surfaces were not always properly cleaned before assembly and bolts were not tightened to the correct torque setting.

All braided flexible connections, laminated connections and bolted joints of generator busbar systems should be inspected for signs of deterioration, overheating or major damage during normal maintenance outages.

Enclosures of main connections can deteriorate due to various mechanisms and require regular inspection whilst in service and during outages.

2.0 Scope:

The document covers the procedure to be followed during the periodical inspection of the braided joints in the generator bus duct of both stg#1,2

3.0 REFERENCE DOCUMENT:

Operation Guidance Note COS-ISO-00-OGN/OPS/ELEC/035 Rev. No.: 1 Aug. 2022

4.0 SUPERSEDED DOCUMENTS:

LMI /OGN/ELEC/035 Issue: 1 Rev : 3 dated : Jan 2022

5.0 BRAIDED FLEXIBLE CONNECTIONS (BFCs)

New braids should be ordered to the requirements of the specification shown in Appendix. Before a set of BFCs (either new or old) is to be fitted the resistance of each braid should be checked with a micro-ohmmeter (with min current 100Amp) and be compared with the average for the whole set. Any braid that is grossly higher in resistance should be rejected.

5.1 Guidance for Fitting BFCs

In addition to selecting the correct BFC as detailed in Appendix it is essential that it is correctly fitted. The following requirements should be met for all new installations. Where, on an existing installation they cannot be met, consideration should be given to its modification.

Braided flexible connectors must not be under tension when fitted and the braid should have 25-35 mm free play when cold.

When fitted the braid must not be twisted or bent edgeways.

The busbars must not protrude beyond the ferrule and thus be liable to come into contact with the braid.

When a number of BFCs are fitted in parallel the braids must be arranged so that they cannot come into contact with each other when the installation is at operating temperature.

When it is necessary to stack braids on top of each other, a 6mm copper spacer will be required between contact surfaces of each BFC. The spacers must not protrude beyond the ferrule and thus be liable to come into contact with the braid.

6.0 JOINTING

The presence of an oxide layer between joint components particularly if aluminum is involved, can result in deterioration and ultimately in failure of the joint. Use of the following equipment and procedure will produce a permanent and satisfactory connection of low resistance. The use of petroleum jelly for outdoor aluminum to aluminum or copper to aluminum joints is not recommended. The same procedure can be used with Shell Ensis or Castrol Rustillo grease. When used outdoors, petroleum jelly applied on the joints may be washed out by rain.

6.1 Equipment

Stainless steel brushes- file and cared type: One brush should be selected and used solely for each type of metal. The brushes should be kept clean and grease free by occasional cleaning in degreasing fluid.

I. Aluminum oxide paper cloth: medium grade.

II. Petroleum jelly (on aluminum/aluminum or copper/aluminum joints, busbar manufacturers may use their own proprietary compound).

III. Torque spanners

IV. Micro ohm meter

V. Degreasing agent for wire brushes and electroplated joint faces.

VI. Joint preparation

VII. Ensure that the joint faces are clean, flat and free from burrs.

6.1.1 Non-Electroplated Joints

Select the correct wire brush for the metal and wire brush contact face. Where large surfaces are to be cleaned such as intermediate busbar joints, a disc sander can be used with aluminum oxide medium grade discs.

Apply a liberal coating of petroleum jelly to each contact face and wire brush each face through the jelly.

Where inaccessibility prohibits the use of a wire brush or file card only the correct abrasive paper/cloth for the particular application should be used. For copper or brass surfaces emery paper is suitable but for aluminum surfaces only an aluminum oxide paper or cloth should be used. Tests have shown that using emery cloth or paper on electrical grade aluminum can leave particles embedded in the surface of the aluminum. Joint faces should be abraded dry and petroleum jelly applied immediately afterwards.

Electroplated Joints

Clean the contact area with a degreasing agent before applying petroleum jelly. If the contact surface shows signs of oxidation this should first be cleaned with a nylon brush before cleaning with a degreasing agent and applying petroleum jelly.

6.2 Joint Assembly

Assemble the joint ensuring that the correct size and arrangement of bolt, nut, lock nut, spring washer, plain washers, spring disc washer or locking nut are used. The arrangement varies with the type and age of the installation, and unless problems have occurred, the original arrangement should be maintained. Joints involving aluminum should always incorporate either a spring disc washer or spring washer. Plain plated washers for aluminum should be as per BS4320, large diameter metric series (or imperial equivalent), thickness as in form C.

Progressively tighten each main nut in turn to the correct torque value. Most aluminum busbars are made from a near pure grade of aluminum, which is relatively soft. Excessive tightening of the bolts at a joint position can result in the aluminum cold flowing, resulting in loss of pressure at the joint.

Recommended bolts, and torque values for aluminum to aluminum or aluminum to copper joints are given for various sizes in Table 1. Many modern aluminum to aluminum bolted joints

use Belleville spring disc washers to maintain the correct pressure. It is imperative that these are renewed if they are found to be damaged or distorted from their original form.

Torque settings for copper to copper or copper to brass joints are dependent on the bolt or stud material type and size.

During maintenance it is important that flexibles are removed from both ends.

If for any reason a joint assembly is slackened it should be completely undone and reassembled after repeating the cleaning and greasing procedure.

6.2.1 Testing of the assembled joint

Where practical, the contact resistance of a bolted joint should be checked using a micro–ohm meter with min current 100A. The joint at one end of a braid can be checked before the other end is fitted, but once multiple parallel paths have been established readings will not be representative of the joint under test. In theory joint resistance should not be higher than the equivalent mean length of unjointed busbar.

7.0 MAIN CONNECTION ENCLOSURES

7.1 Isolated Phase Busbar Trunking (IPBT) System

The outer trunking sheath continuity between sheath bonding position must be always maintained. Should a sheath become open circuited between bonding positions, high flux levels created by the current in the main conductor will not be cancelled by the absent sheath current, leading to adverse effects on adjacent metalwork and equipment.

At enclosed current transformer positions where the access covers form part of the sheath it is imperative that after removal for inspection etc. the cover joints are remade to ensure that a good continuity is maintained.

The trunking sheath mounting points to support steelwork are insulated to ensure that sheath currents are contained in the sheath and do not circulate in random paths. The sheath system is then connected to each at one point. Support insulation of earlier installations was rather basic, consisting of sleeved bolts and insulation shims between mounting surfaces. Later installations were much improved.

The insulation of the sheath system to earth should be checked at major outages and if anything is suspected, it may be replaced. Infra-red camera surveys should also be undertaken at convenient times to ensure no excessive temperatures are developing on the sheath system.

Most IPB trunking enclosures have dehumidifier equipment. It is imperative that this is operating correctly, especially when a unit is returning to service after a long outage, where condensation may have accumulated in the IPB system whilst the dehumidifier has been out of service. In Generator Busducts where line side adopter chamber is bolted with the generator and mesh with cover provided for hydrogen leakage escape, checking for any blockage in wire mesh must be done.

7.2 Enclosed system

Older main connection busbar systems were enclosed in fabricated steel and/or aluminum housings. Some have interphase barriers of either aluminum or insulation material. The housings can be subject to overheating at various points due to circulating or induced currents and there have been instances of fire when leaking turbine or transformer oil and other combustible materials have come into contact with these areas. Most problems have arisen with steel framed aluminum clad systems, where deterioration of insulated joints in the frame structure has resulted in circulating current and subsequent overheating at bolted joint positions. Older installations used paper or fabric-based insulation shims and bolt sleeving at these points. Wear due to vibration, removal and incorrect refitting can lead to poor joints. Epoxy glass insulation has proven to be more reliable and is not easily damaged Aluminum structures can suffer from overheating at bolted joints where the aluminum joint resistance is high. As a general rule joints should either be properly insulated or made in such a manner that their electrical resistance is as low as possible.

Most overheating areas of enclosures can be detected externally by a comprehensive thermal camera survey before they become a problem. Enclosures, their fittings and internals should be inspected during major outages for deterioration.

8.0 SPECIAL INSTRUCTIONS TO BE FOLLOWED:

In stage#1, there are total 8 places where braided connections are present.

Similarly in Stage#2, there are 4 places where braided connections are present.

SLNO	LOCATION	STAGE#1	STAGE#2	RESPONSIBILITY
1	Gen and 21 KV bus duct	√	√	Area Engineer – Gen Group
2	GCB Incoming	√	-	Area Engineer – Switchyard Group
3	GCB Outgoing	√	-	Area Engineer – Switchyard Group
4	Expansion bellow @ TG A-row	√	-	Area Engineer – Switchyard Group
5	21 KV VT	√	√	Area Engineer – Switchyard Group

6	DUCT AND GT	√	√	Area Engineer – Transformer Group.
7	DUCT AND UT-A	√	√	Area Engineer – Transformer Group.
8	DUCT AND UT-B	√	-	Area Engineer – Transformer Group.

Thermo vision scanning of all the 21 KV generator bus duct joints at bellows will be done once in month as per the PDM schedule.

Instruction	Responsibility	Document Ref.
Thermo vision scanning of bus duct joints as per PDM schedule	Area Engineer	LMI /OGN/ELEC/035

Dew point measurement of bus duct pressurizing air once in a month.

Instruction	Responsibility	Document Ref.
Dew point measurement of air in the bus duct pressurization unit	Area Engineer	LMI /OGN/ELEC/035

9.0 METHODOLOGY OF IMPLEMENTATION:

Temperature of all the joints in the 21 KV generator bus duct are to be monitored by thermo vision scanning on Monthly basis.

Temperature above ambient	Action Plan
> 150°C	Immediate
100°C – 150°C	During next opportunity of Shut down
80°C – 100°C	Monitoring the trend

All the joints are to be physically inspected during unit overhauls.
If any damages / marks are observed on the flexible, immediately it has to be replaced.
Tightness of the joint is to be ensured with torque wrench as per table 1 given below.

10.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION:

Generator bus duct maintenance area engineer and transformer maintenance engineer for implementation and overall responsibility lies with sectional heads of electrical maintenance.

11.0 EXCEPTION REPORTING AND DISPENSATION OF DEVIATIONS

Any exceptions / deviations shall be reported to HOD (EM).

12.0 RECORD KEEPING:

1. Healthiness of Flexible connections as per Annexure-1
2. Thermo vision scanning data
3. Dew Point measurement data

13.0 REVIEW :

This LMI document will be reviewed once in every two years by Head ofthe Project, NTPC Simhadri.

APPENDIX: SPECIFICATION FOR THE DESIGN AND MANUFACTURE OF BRAIDED FLEXIBLE CONNECTORS (COPPER)

A.1 INTRODUCTION

This Specification is intended to cover the basic principles to be adhered to in the design and manufacture of copper braided flexible connectors (BFC) of nominal current rating greater than 50 amp.

It is envisaged that BFCs used in power station apparatus (in particular generator main connections), switchyards and similar or associated apparatus will be supplied in line with this specification in order to achieve a high standard of design, manufacture and reliability necessary in such areas.

A.2 INFORMATION REQUIRED TO SELECT A BFC

The following information including a drawing, should be given to the BFC Manufacturer to enable him to advise on selection of the appropriate BFC.

I. Overall length of the braided flexible connector
II. Compressed ferrule length
III. Compressed ferrule width
IV. The number, size and position for fixing holes
V. Full details of the installation into which the BFCs are to be fitted
VI. Air temperature within the enclosure
VII. Terminal temperatures (positive or negative heat sinking).
VIII. Number of connectors in parallel
IX. Configuration of terminals

Note that the selection of the correct current rating for the completed flexible connector should use Table II for guidance only. The braids should be capable of operating continuously at a temperature of 90°C unless otherwise specified.

The braid manufacturer has responsibility for: -

Manufacture to the specification if the purchaser defines the detailed design of the BFC.

Manufacture to specification and suitability for application if the purchaser defines the braid duty only.

A.3 DEFINITIONS

For the purpose of this document the terms used throughout are defined illustratively. It should be noted that these terms are not claimed to be universally definitive (i.e., different manufacturers may use different terminology) but should be generally understood.

A.4 DESIGN AND MANUFACTURE

The BFC is to be manufactured from braided tinned copper wire and is to have crimped copper ferrules as described below:

A.4.1 Copper Braid

The copper braid shall be manufactured from tinned copper wire as per BS EN 13602.The individual wire strand diameter shall be from 32 to 38 SWG. All strands shall of be the same nominal diameter.

Table II shows the approved braid constructions. Table III shows their associated current/temperature characteristics.

These constructions must be used wherever possible. Type tests must be performed as specified in section A.7. Alternatively, proof of previous type tests must be made available to the purchaser. Type tests must be carried out if alternative construction is to be used.

A.4.2 Copper Ferrule Tubes

Copper ferrule tubes must have a thickness (excluding tinning) before crimping of not less than
3.25 mm and should be suitable to withstand the pressure applied during manufacture. The crimping pressure shall not be less than 40 kg/mm over the area on which pressure is applied. The ferrule tube is to be hot tin dipped (other processes are not acceptable). The tinning to be carried out using soft solder, 60/40 (tin/lead), as per BS EN 29453. The coating thickness at any point is to lie between the limits. 0.005 mm and 0.013 m. It is important that the inside surface of the ferrule tube is fully tinned and checked by visual inspection.

A.4.3 Compressed Ferrule

Compressed ferrules with thickness greater than 10 mm are required to have a crimped side face. The crimp must deform the braid when viewed from the end and not be a “cosmetic” crimp which only deforms the ferrule tube.

A.4.4 Completed BFC

It is important that: -

I) All strands run the full length of the ferrule

II) The braid has the correct construction

III) The braid material is tightly packed i.e. additional filler material must not be used within the ferrule.

IV) Solder has not been used as a filler

After crimping, all contact faces are to be polished to give a natural copper finish. The contact surface faces shall be left with a surface flatness of 0.1 mm or better over their length and 0.05mm or better over their width. The polishing process shall not unduly reduce the thickness of the ferrule tube and at no point shall the finished ferrule tube have a thickness less than 90% of the original ferrule tube.

A.5 TYPE TESTING

When a manufacturer proposes to make type tests, he shall notify the purchaser of the programme and procedure in writing. After approval of the programme and procedure adequate notice of such tests should be given so that arrangements can be made for the tests to be witnessed. The tests shall include routine tests A.6.1 to A.6.3 as well as those listed below. A test certificate shall then be issued by the manufacturer for the BFC construction and braid make up.

A.5.1 Braid Only

Type tests must be carried out to establish the current/temperature characteristics of both single and multiple layer arrangements of braided material. The tests shall be carried out within a sealed aluminum box having dimensions sufficient to ensure that the temperature rises at the centre of the test sample are unaffected by end effects where the sample passes through the box.

A.5.2 Completed BFC

As part of type testing of a BFC a sample must be supplied and cut open for inspection as detailed in clause A.6.2.

A.6 ROUTINE TESTS

The following routine tests shall be performed, and an appropriate test certificate issued for each batch as defined below:

A.6.1 Non-Destructive Test on Ferrule Tube Only

This test is to be carried out on a randomly chosen sample as defined below on ferrule tubes for each size of BFC covered on any one order.

1 sample for up to 10 ferrule tubes

2 samples for 11-20 ferrule tubes

3 samples for 21 and above ferrule tubes

The samples shall be taken in the condition in which they would be found immediately prior to crimping and shall be checked for dimensions and thickness against the manufacturing specification. Tinning thickness Shall be measured using a metallurgical microscope using one of the methods described in BS1872.

A.6.2 Destructive Test on Completed BFCs (at the option of the Purchaser)

In order to ensure compliance with section A.4.4 the Purchaser, if specified at time of order, is at liberty to select a sample which shall be broken open by cutting along the length of the ferrule in a plane bisecting the side faces. The braid shall then be removed from the ferrule and checked that it conforms to specification.

A.6.3 Tests on BFC

All completed braided flexible connectors shall be subjected to the following routine tests.

A.6.3.1 Visual and dimensional check against the original manufacturingdrawing checking for:

I. Number of layers

II. Free length of overall connector

III. Position and dimension of fastener holes

IV. Dimensions of contact faces and flatness.

V. General Quality and Finish

A.6.3.2 A test to determine the resistance of the completed connector

The test method and criteria are as given in section A.7. Any connectors which fail this test shall be rejected and clearly and indelibly marked as such at the time of the test. If more than 10% of the original number ordered of each different type of BFC fail the resistance test the cause of and solution to the problem shall be the subject of discussion between the parties concerned.

A.7 BRAID PERFORMANCE TYPE TEST

A.7.1 Test Description

The tests shall be carried out in a purpose-built enclosure to maintain draught free conditions, constructed of aluminum (or similar) having dimensions of approx. 2.4 M X 1.2 M X 1.2M and with its inside painted black.

The braid shall enter and leave the enclosure via electrically insulated holes allowing negligible air flow and be positioned along the central axis of the enclosure. The braid shall be clamped to terminal blocks/busbar outside the enclosure and shall be supplied with 50Hz AC from a suitable low voltage, high current controllable source. Thermocouples shall be mounted along the braid in the enclosure at five equal spaced positions and at each end external to the enclosure. Thermocouples shall be placed to measure the air temperature in the enclosure and the external surface temperature at the sides and top. When there is more than one layer, the thermocouples should be fitted on the top braid only. The current in the braid shall be measured to an accuracy better than 2% and the temperature to an accuracy of $\pm$ 0.5deg C.

A.7.2 Test Sample

The braid test samples shall comprise braid of the same construction as to be used in the layers of the production of BFC. Tests shall be performed on a single layer and all numbers of layers up to and including the number to be used on the production BFC. When more than one layer is being tested the layers shall be wired firmly together using tinned copper wire at intervals throughout the sample length so as to prevent the layers sagging apart.

The sample should be clamped firmly to the terminal blocks such that it is neither under undue tension or sagging excessively.

A.7.3 Test Method

Immediately prior to a test the ambient temperature inside the enclosure shall be measured and recorded. Various current shall be passed through the braid to obtain temperature increases in steps of approximately 20° C up to a maximum detected braid temperature of approximately 105° C reached by any one braid thermocouples. At each step the current shall not be increased until the temperature rise has stabilized at which time all temperatures and the current shall be recorded. The temperature distribution along the braid shall be uniform and there shall be no instance of over-heating at joint interfaces or within the braid.

A.7.4 Presentation of Results

This sequence of events shall be presented in both tabular and graphical form for each braid. The graph shall illustrate the various current/temperature characteristics for the various numbers of layers together with the average of the ambient temperatures measured inside the enclosure. The graph and tables must also clearly identify the braid construction. This data shall form the basis of the justification for the claimed rating for the completed BFC.

Once type testing has been successfully completed no change in construction is permitted without approval of the purchaser.

A.8 RESISTANCE TEST

This routine test, to be performed on all BFCs is a resistance measurement. The measured resistance must not exceed a value of calculated maximum resistance determined from an effective resistivity of the connector of 25.0 micro-ohm/ mm. This figure was established from tests carried out by CERL on a large number of connectors.

The detailed design of the apparatus in terms of jigs etc., to facilitate this routine test process, is left to the manufacturer. While doing the test, it may be seen that test current from a suitable low resistance is passed between symmetrically placed probes on the contact faces. The probe shall have a contact face of at least the width of the ferrule. The voltage drop measuring probes shall have hard, pointed tips and shall be applied with sufficient force to penetrate any minor pollution that may be present on the contact face but shall not damage the contact face in any way.

A.8.1 Test Procedure

The temperature of each sample should, if possible, be controlled in the region of 18-20° C and shall be recorded immediately prior to the commencement of the tests (temperature T).

The tests shall made using a low resistance test set having an accuracy of at least 0.1A reading, +0.2 micro-ohm; at 20° C.

A.8.2 Resistance Calculation of The Connector

An initial calculation shall be made to find the maximum allowable resistance (Rmax) at 20° C using the formula:

$R_{max} = (25.0 \times L)/CSA$

Where CSA is the actual cross section in mm2 of the metal conductors in the braid calculated from the number of wire strands and their individual cross-sectional area.

L is the length in mm between the voltage test points, without stretching or compression of the BFC.

A.8.3 Pass/Fail Criteria

The measured resistance RT of the connector shall be corrected to a resistance R20 at 20° C by the following formula:

$R_{20} = R_T \{1+0.004(20-T)\}$

Resistance R20 shall not exceed resistance Rmax.

TABLE I

TORQUE SETTINGS FOR STEEL BOLTS USED ON ALUMINIUM JOINTS

Bolt Size	Torque Setting	
	Nm	lbs ft
M16	88	65
M12	66	45
M10	41	30
M8	15	11
M6	7	5

TABLE II

BRAID CONSTRUCTIONS

Braid Coding	Nominal Braid Width mm	Specification Coats x Bobbins/No. of Wires per bobbin & Wire gauge	Conducting CSA mm
1	48	7x48/20/38 SWG	122
2	51	5x48/22/38 SWG	95
3	40	1x48/24/32 SWG	68
4	19	2x24/29/38 SWG	25
5	38	2x48/20/32 SWG	113
6	86	3x48/25/32 SWG	212
7	152.5	3x48/25/32 SWG	212

TABLE III

CURRENT/TEMPERATURE CHARACTERISTICS OF BRAID

Current Resulting in Braid Temperature Rises of 50° C (British Specification) and 65° C(American Specification) in an Ambient of 20-25° C.

Braid Type Number	Number of Layers	Current (in Amps) Required to obtain temperatures of	
		50° C Rise	65°C Rise
1	1	425	490
	2	660	780
	3	890	1035
	4	1060	1235
2	1	375	430
3	1	310	350
4	1	135	155
5	1	395	445
6	1	720	830
	2	1045	1185
7	1	770	885

ANNEXURE:1 JOINT CHECKLIST FOR IPBT BUS DUCT FLEXIBLE CONNECTORS INEPECTION (DURINGO/H)

Activity	Condition (OK/NOT OK)	Responsibility	Any remarks
Connector's Bolt Tightness checking at complete IPBT Bus duct sections & GT/UT Bushings as per standard torque.		EMD & FQA	
Connectors physical condition checking & cleaning		EMD & FQA	
Healthiness checking of connectors ferrules.		EMD	
Resistance measurement of each connector (Record to be maintained & compare with last result)		EMD & FQA	

Signature of EMD Representative Signature of FQA Representative